Chapter 3- Computer Ports

Parallel Port

A **parallel port** is used to transfer data between a computer and peripheral devices (like printers) using a **25-pin or 36-pin connector**.

Key Features

- **Parallel Communication**
  Multiple bits (usually 8 bits = 1 byte) are transmitted **simultaneously** over multiple wires.
- **IEEE 1284 Standard**
  With the IEEE 1284, the parallel port became:
  - **Bidirectional**
  - Able to **send and receive data**

Speed

- Basic speed: **50 KB/s to 150 KB/s**
- Advanced modes: Up to **2 MB/s**

Modes for High Speed

- **EPP (Enhanced Parallel Port)**
  Faster data transfer with low CPU usage.
- **ECP (Extended Capabilities Port)**
  Supports:
  - Data compression
  - DMA (Direct Memory Access)

These modes use:

- 8-bit hardware interface
- Internally operate like 16-bit or 32-bit transfer systems

Serial Port

A serial port is an electronic communication interface used to transfer data one bit at a time between a computer and peripheral devices.

Key Features

- **Serial Communication**
  Data is transmitted bit by bit sequentially over a single channel/wire.
- **Common Usage (Past)**
  Along with the parallel port, serial ports were widely used for:
  - Modems
  - Keyboards
  - Mice
  - Printers

Modern Replacement

Today, traditional serial ports have mostly been replaced by faster and more versatile interfaces such as:

- USB → General-purpose high-speed data transfer
- Ethernet → Networking and internet communication
- VGA → Display/video output

Why Serial Became Popular (and Still Exists)

- Requires fewer wires than parallel communication
- More reliable over long distances
- Less signal interference

PS2 Port

The PS2 port is a 6-pin mini-DIN connector used to connect keyboards and mice to a computer.

Features of PS2 Port

Dedicated port: PS2 ports are dedicated, meaning one port is used only for a keyboard and another only for a mouse. They cannot be interchanged.

Connector type: It uses a 6-pin mini-DIN connector.

Color coding:

Purple: Keyboard

Green: Mouse

Direct communication: The PS2 port allows devices to communicate directly with the motherboard without needing complex drivers.

Speed and capability: PS2 ports provide reliable and fast communication, especially for keyboards, with minimal delay.

Interrupt-based: It uses hardware interrupts, so the CPU is immediately notified when a key is pressed or mouse is moved.

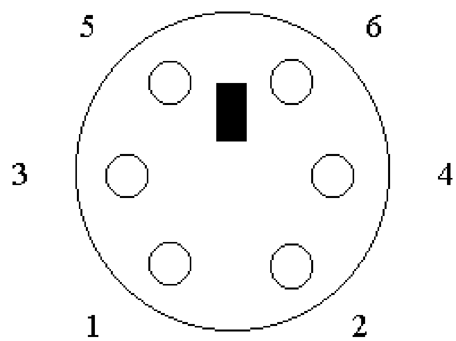
Purpose

The main purpose of the PS2 port was to replace serial ports used for keyboards and mice. It provided better performance and simpler connection.

Modern usage

Nowadays, the PS2 port has mostly been replaced by USB ports because USB is more flexible, supports many types of devices, and is easier to use. However, PS2 ports are still preferred in some cases, especially for keyboards, because they offer faster response and support full key rollover.

PS/2 Keyboard and Mouse Cable



Pin	Name
1	+Keyboard Data
2	Unused
3	Ground
4	+5 Volts
5	Clock
6	Unused

Cable (male) pinout

USB Port (Universal Serial Bus)

The USB (Universal Serial Bus) port is the most widely used interface for connecting peripheral devices such as keyboards, mice, printers, flash drives, and external storage to a computer. It was developed to provide a simple, fast, and standardized method for data transfer and power supply between devices. USB is maintained by the USB Implementers Forum.

Features of USB Port

Universal connectivity: Supports many types of devices like keyboard, mouse, printer, scanner, and storage devices.

High data transfer speed:

USB 2.0 → up to **480 Mbps**

USB 3.0 → up to **5 Gbps** (bi-directional)

Power supply: USB can supply power to devices, so many devices do not need external power.

Supports multiple devices: Using a USB hub, up to **127 devices** can be connected to one USB port.

Plug and Play: Devices are automatically detected and configured.

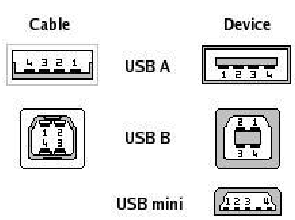
No clock signal required: Uses its own synchronization method.

4 pins (USB 2.0):

- 2 for data transfer
- 1 for power
- 1 for ground

USB 2.0 - 4 Pins and their functions

- **VCC (Pin 1)** → Supplies **+5V power** to the device
- **D− (Pin 2)** → Used for **data transfer (negative signal)**
- **D+ (Pin 3)** → Used for **data transfer (positive signal)**
- **GND (Pin 4)** → **Ground** connection



Pin	Signal	Color	Description
1	VCC		+5V
2	D-		Data -
3	D+		Data +
4	GND		Ground

Types of USB Connectors

1. USB Type-A

- Most common USB connector
- Found on computers, laptops, and chargers
- Used for connecting flash drives, keyboards, and mice

2. USB Type-B

- Square-shaped connector
- Used mainly for printers and scanners

3. USB Micro-A

- Smaller and thinner than Mini USB
- Used in some mobile and portable devices

4. USB Micro-B

- Very common in smartphones, power banks, and portable devices

5. USB Mini-A

- Smaller version of USB-A
- Used in older portable devices

6. USB Mini-B

Common in older cameras, MP3 players, and external devices

Why is PS2 considered better than USB?

PS2 is better than USB in data transmission because it uses a dedicated port and interrupt-based communication. It does not share bandwidth with other devices and sends data directly to the CPU without polling. This makes PS2 faster and more reliable for keyboard and mouse input.

Why has USB replaced the PS2 port?

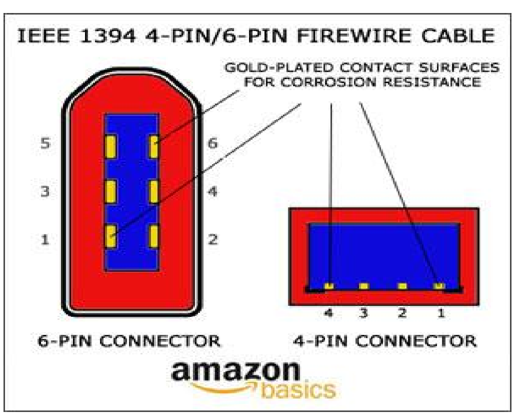
PS2 was replaced by USB because USB is more flexible and convenient to use. If a PS2 device is unplugged and plugged back in, the computer usually needs to be restarted to detect it. In contrast, USB supports hot-swapping, so devices can be connected or removed without restarting the system. USB also supports more types of devices, provides power, and is a faster and improved version of the serial port.

USB reversible:

A reversible USB (like USB-C) can be plugged in either way- upside down or right side up- because of its symmetrical design. It's easier to use, supports faster data transfer, higher power, and can carry data, video, and power in one cable.

FireWire Port (IEEE 1394)

The FireWire port, also called **IEEE 1394**, is a high-speed interface used to connect multiple devices such as external hard drives, digital cameras, and audio/video equipment. It is designed for **fast and reliable data transfer**.



Versions and Speeds

FireWire 400 – Original standard.

FireWire 800, 1600, 3200 – Later versions offering much higher data transfer speeds compared to FireWire 400.

Features

Multiple Device Connectivity: Multiple devices can be connected using a **FireWire hub**.

Plug and Play Support: Devices are automatically recognized without manual configuration.

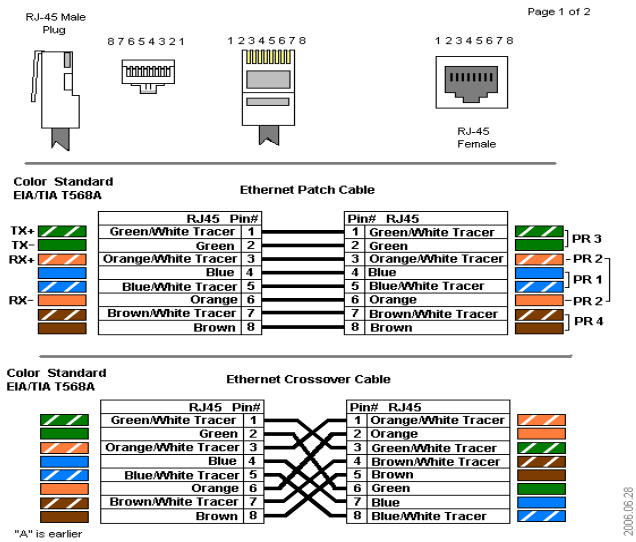
Operating System Compatibility

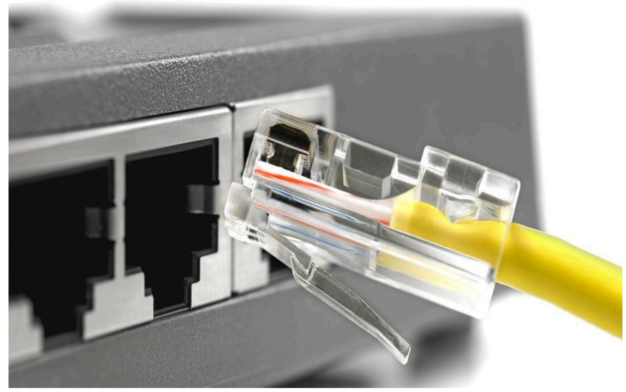
The FireWire port supports a wide range of operating systems, including:

- Mac OS 8.6, 9, Mac OS X
- Microsoft Windows
- Haiku, Linux, FreeBSD, NetBSD

Ethernet Port

An Ethernet port is a network interface used for data transfer and communication between devices supporting the IP protocol. It is the standard interface for wired local area networks (LANs).





Cabling

Twisted Pair Cable: Used for **short distances** (computer to switch or router).

Coaxial Cable: Used in some older Ethernet setups.

Fiber Optic Cable: Preferred for **long distances**, providing higher speed and minimal signal loss.

Cable Length and Signal Considerations

Maximum length for twisted pair Ethernet cable is 100 meters (328 ft); beyond this, signal loss and connection issues may occur.

For longer distances, **signal boosters** can be installed every 100 meters.

Fiber optic cables are more efficient for long distances and, with proper equipment, can transmit data up to **20 km** without boosters.

Speed

Ethernet speeds range from **10 Mbps to 10 Gbps**.

High-speed connections are commonly used in **large networks** (e.g., CCTV IP networks or data centers).

In many setups, high speed occurs **between switches or routers**, while end users may receive **10 Mbps or 100 Mbps**.

Applications

- Local Area Networks (LANs)
- Data centers and high-speed network backbones
- CCTV/IP surveillance networks
- Internet and intranet connectivity

Q. What types of cables are required to connect an ISP to a home, and what are their purposes?

- **Fiber optic** brings internet from the ISP to our home for speed and long-distance reliability.
- **Ethernet cable** distributes that internet inside our home to computers and devices.

Difference Between Ethernet and Fiber Optic Cable:

Feature	**Ethernet Cable (Twisted Pair)**	**Fiber Optic Cable**
Transmission Medium	Copper wires	Glass or plastic fibers
Distance Limit	Up to **100 meters** per segment	Up to **20 km** or more with proper equipment
Speed	10 Mbps – 10 Gbps	100 Mbps – 100 Gbps+

Signal Loss	Significant loss over long distances; may require **signal boosters** every 100 meters	Minimal signal loss; usually no boosters needed
Interference	Susceptible to electromagnetic interference (EMI)	Immune to EMI
Cost	Lower cost	Higher cost
Use Case	Short-distance LAN connections (home, office)	Long-distance internet backbone, FTTH, high-speed networks

Need for Signal Boosters in Ethernet

Ethernet cables lose signal strength beyond **100 meters (328 ft)**. To maintain connectivity over longer distances, **signal boosters or repeaters** must be installed at intervals.

Fiber optic cables, on the other hand, can span **much longer distances without boosters**, making them ideal for ISP-to-home connections.

VGA and HDMI Ports

VGA (Video Graphics Array) and HDMI (High-Definition Multimedia Interface) are standards used to connect devices, such as laptops and DVD players, to displays like TVs, monitors, or projectors.

Signal Support

VGA: Carries **only video signals**.

HDMI: Carries **digital video and audio signals**, and supports **HDCP encryption**.

Capabilities

VGA: Can transmit **1 channel for video**; other signals require separate cables.

HDMI: Can transmit **1 channel for video, 8 channels for audio**, and CEC communication between devices using a single cable. It also **encrypts data using HDCP**.

Connectors

VGA: Uses a **15-pin connector**, usually marked blue.

HDMI: Several types exist (Type A–E); the most common is **HDMI Type A** with **19 pins**.

Video Quality

VGA: Maximum resolution **2048 × 1536**, but may have ghosting or visual issues.

HDMI: Supports a wide range of resolutions **without signal loss or interference**.

Additional Features

HDMI: More stable, carries audio, and can carry internet data in newer versions.

VGA: Does not support audio or digital encryption.

Summary:

1.VGA is an analog video standard while HDMI is in digital
2.VGA is very old while HDMI is still pretty new
3.VGA can only carry a video signal while HDMI can carry a lot of other signals along with video
4.VGA is usually used in computers while HDMI is used in HD tv sets and media players

1) Which one is better to watch movies?

HDMI is better because:

- Supports high-definition video without signal loss.
- Carries digital audio along with video, so no need for extra cables.
- Supports HDCP encryption, required for Blu-Ray and streaming content.

VGA only carries video and may show ghosting or lower quality at high resolutions.

2) Which One is Better for Gaming?

HDMI is better for gaming because:

- Supports high resolutions and refresh rates, giving smoother visuals.
- Digital signal ensures no interference or loss, unlike analog VGA.
- Can carry audio to speakers/headsets, reducing cable clutter.

VGA is outdated for gaming, especially on modern HD/4K monitors, because it cannot match HDMI's clarity and stability.

3)Why HDMI is Better Than VGA?

Carries both video and audio: HDMI transmits 1 video channel and 8 audio channels, while VGA carries only video.

Higher video quality: HDMI supports higher resolutions without ghosting or interference; VGA quality degrades at high resolutions.

Data encryption: HDMI encrypts data using HDCP; VGA does not.

Single cable convenience: HDMI can handle video, audio, and device communication (CEC) with one cable; VGA requires separate cables.

Difference Between VGA and HDMI

Feature	VGA	HDMI
Channels	Can transmit 1 channel for video only	Can transmit 1 channel for video, 8 channels for audio, and CEC communication
Signal Type	Carries analog video only	Carries digital video and audio
Encryption	No encryption is supported	Supports HDCP encryption for secure data
Resolution	Supports lower resolution; may require boosters for long cables	Supports high resolution without signal loss